

# IDS Group Project Data Setting & Cleaning

K9

2025-12-01

## Assignment brief

- Perform exploratory data analysis to assess how 6 continents in the world (all continents except Antarctica) are faring with respect to the following specific targets set out by the UN:
  1. Sustain per capita economic growth in accordance with national circumstances and, in particular, at least 7 per cent gross domestic product growth per annum in the least developed countries.
  2. By 2020, substantially reduce the proportion of youth not in employment, education or training.
- Find another CSV file to add to analysis

*(Please note that hereafter education, employment, and training would be referred to as EET)*

In this chunk, we would import the data.

```
continents <- read.csv("data_sets/continents-according-to-our-world-in-data.csv")
gdp_per_capita_df <- read.csv("data_sets/gdp-per-capita-worldbank.csv")
youth_not_in_eet <- read.csv("data_sets/youth-not-in-education-employment-training.csv")

# We now create a vector of all LDC countries
ldc_countries <- c(
  "Afghanistan", "Angola", "Bangladesh", "Benin", "Bhutan", "Burkina Faso",
  "Burundi", "Cambodia", "Central African Republic", "Chad", "Comoros",
  "Democratic Republic of the Congo", "Djibouti", "Eritrea", "Ethiopia",
  "Gambia", "Guinea", "Guinea-Bissau", "Haiti", "Kiribati",
  "Lao People's Democratic Republic", "Lesotho", "Liberia", "Madagascar",
  "Malawi", "Mali", "Mauritania", "Mozambique", "Myanmar", "Nepal",
  "Niger", "Rwanda", "Sao Tome and Principe", "Senegal", "Sierra Leone",
  "Solomon Islands", "Somalia", "South Sudan", "Sudan", "Timor-Leste",
  "Togo", "Tuvalu", "Uganda", "United Republic of Tanzania",
  "Vanuatu", "Yemen", "Zambia"
)

population <- read_csv("data_sets/population.csv", skip = 3)
```

We now clean the data, getting rid of any irrelevant data.

```
# The year is irrelevant in continents, as the only thing this df needs to be
# used for is assigning a continent to a country
continents$Year <- NULL
continents$Entity <- NULL

# We now assign the continent to each country in gdp per capita
gdp_per_capita_df <- gdp_per_capita_df %>%
  left_join(continents, by = "Code")
```

```

# Same thing for youth not in eet
youth_not_in_eet <- youth_not_in_eet %>%
  left_join(continents, by = "Code")

# The continents table is now redundant (Commented this out as when running
# this code seperately without the previous chunk it would produce error)

# rm(continents)

# We clean population data

population <- population %>%
  # Remove indicator code&name, and the last column which is all NAs
  select(-`Indicator Code`, -`Indicator Name`, -70) %>%
  rename(
    Entity = `Country Name`,
    Code = `Country Code`
  ) %>%
  pivot_longer(
    cols = 3:67,
    names_to = "Year",
    values_to = "Population"
  )

```

Still cleaning (renaming columns to nicer names).

```

# Rename a column in gdp per capita
gdp_per_capita_df <- gdp_per_capita_df %>%
  rename(Gdp_per_capita = GDP.per.capita..PPP..constant.2017.international...)

# Rename a column in youth not in eet
youth_not_in_eet <- youth_not_in_eet %>%
  rename(
    Perc_youth_not_in_eet =
      Share.of.youth.not.in.education..employment.or.training..total....of.youth.population.
  )

```

Dropping values before the year 2014, and if population data is above 2021, and if continent is NA.

```

# Drop all values before 2014, as these UN goals were set in 2015
gdp_per_capita_df <- gdp_per_capita_df %>%
  filter(Year >= 2014 & !is.na(Continent))

youth_not_in_eet <- youth_not_in_eet %>%
  filter(Year >= 2014 & !is.na(Continent))

```

Calculating Economic Growth and change in eet of each country

```

gdp_per_capita_df <- gdp_per_capita_df %>%
  mutate(Growth = ifelse(lag(Entity) != Entity, NA, (Gdp_per_capita - lag(Gdp_per_capita)) /
    lag(Gdp_per_capita) * 100))

youth_not_in_eet <- youth_not_in_eet %>%
  mutate(Perc_change = ifelse(lag(Entity) != Entity, NA, (Perc_youth_not_in_eet -
    lag(Perc_youth_not_in_eet)) / lag(Perc_youth_not_in_eet) * 100))

```

This chunk is now to add a flag to the gdp per capita dataframe, labelling if a country is an LDC or not

```
gdp_per_capita_df <- gdp_per_capita_df %>%  
  mutate(Ldc = Entity %in% ldc_countries)
```

Merging the data frames to include population in both

```
gdp_per_capita_df <- merge(gdp_per_capita_df, population, by = c("Entity", "Year"))  
youth_not_in_eet <- merge(youth_not_in_eet, population, by = c("Entity", "Year"))
```

Now we convert to numeric and calculate GDP per capita of each country each year

```
gdp_per_capita_df$Population <- as.numeric(gdp_per_capita_df$Population)  
  
gdp_per_capita_df <- gdp_per_capita_df %>%  
  mutate(Gdp = Population * Gdp_per_capita)
```

Finally, export the CSV files of the 2 data frames

```
write.csv(gdp_per_capita_df, "Exported_data/gdp_per_capita_df.csv", row.names = FALSE)  
write.csv(youth_not_in_eet, "Exported_data/youth_not_in_eet.csv", row.names = FALSE)
```

## Reference

World Bank (2025). Population, Total. [online] The World Bank. Available at: <https://data.worldbank.org/indicator/SP.POP.TOTL> [Accessed 1 Dec. 2025].

## LLM

25/11/2025 asked ChatGPT “what would the definition of “least developed countries” mean in terms of gdp per capita (PPP)”

26/11/2025 asked ChatGPT “give me a list of all countries the UN considers LDCs between 2015 and 2021, with each country in quotation marks seperated by commas”